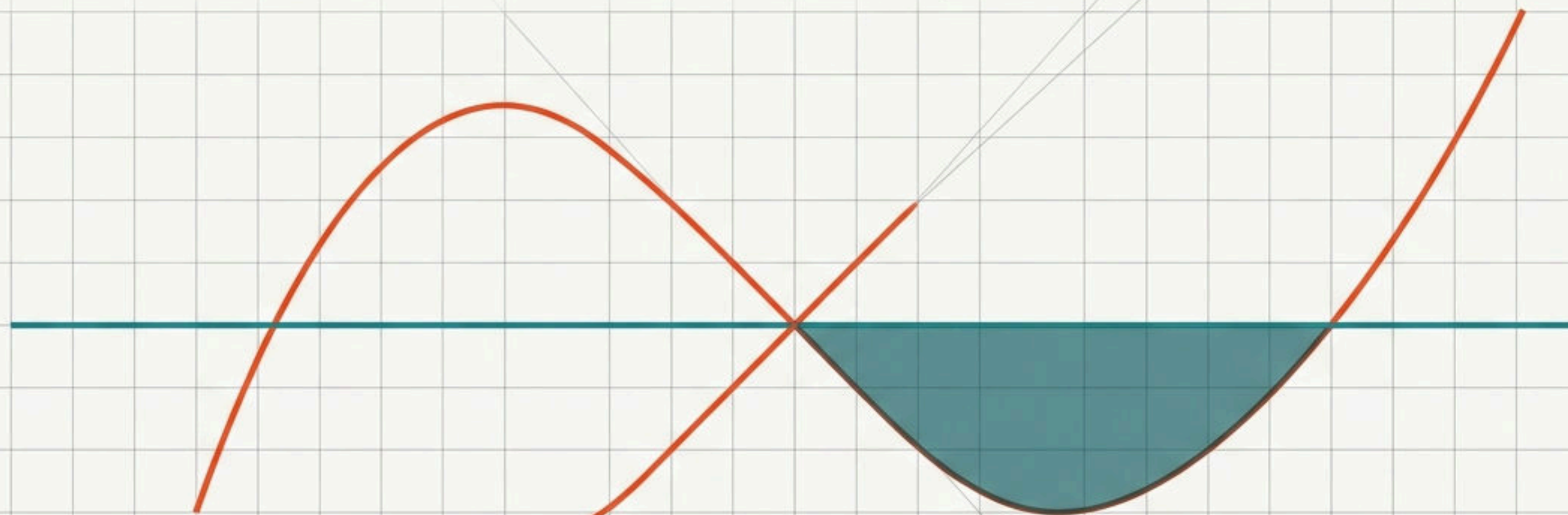


The Threshold of Deferral

A Crossover Theorem for Maintenance and Repair

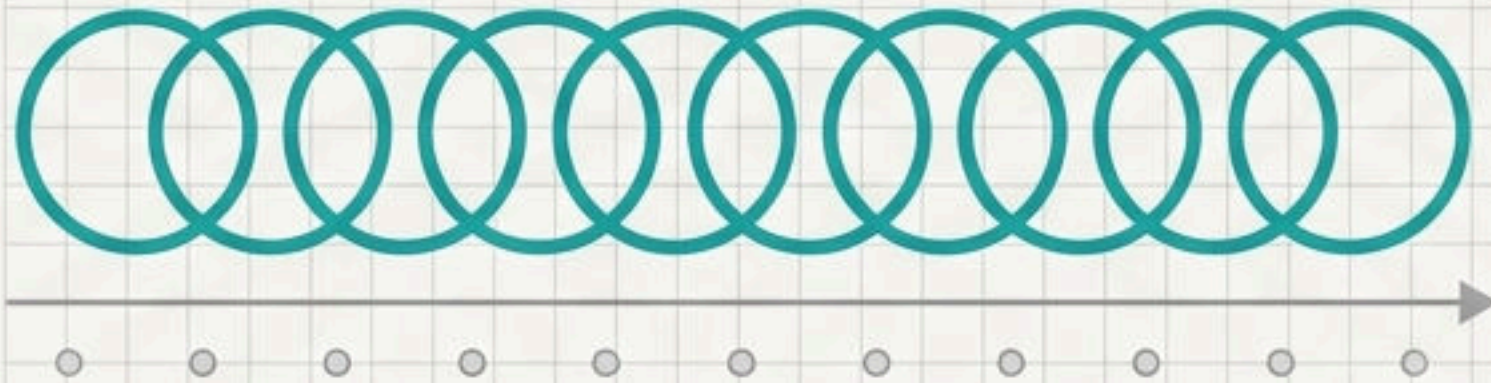


Flyxion, 2026

An executive briefing on the mathematical regimes of system correction.

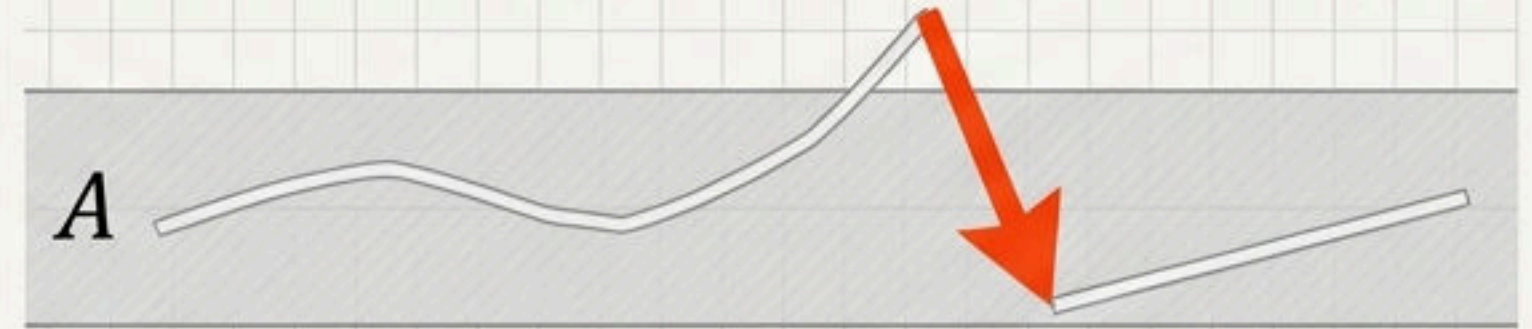
Two distinct kinds of correction

Maintenance (M_t)
Unconditional



Acts on every step to reduce deviation before it compounds.

Repair (ρ_t)
Conditional



Triggered only when the system state (x_t) leaves the viability manifold.

The core question is not whether maintenance is preferable, but under what exact conditions a rational system switches from one to the other.

The dynamics of uncorrected deviation.

Equation Anatomy

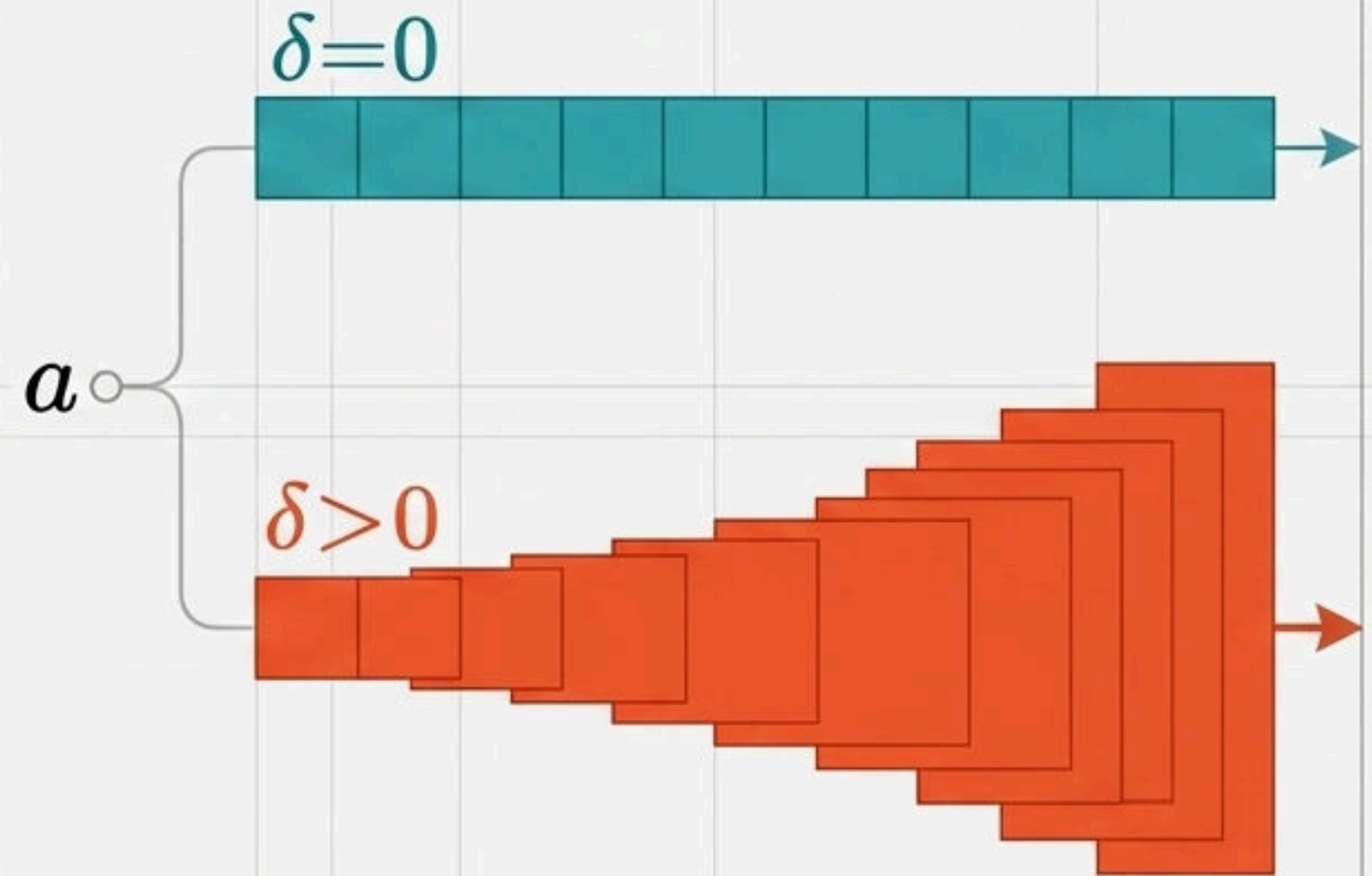
$$d_{t+1} = (1 + \delta)(d_t - m_t) + a$$

a : Deviation introduced at each step.

m_t : Maintenance applied before compounding.

δ : The delay-amplification rate.

Branching Growth Visual



$\delta = 0$ is a distinct regime of pure accumulation without active amplification (e.g., dormant defects).

The competing cost functionals.

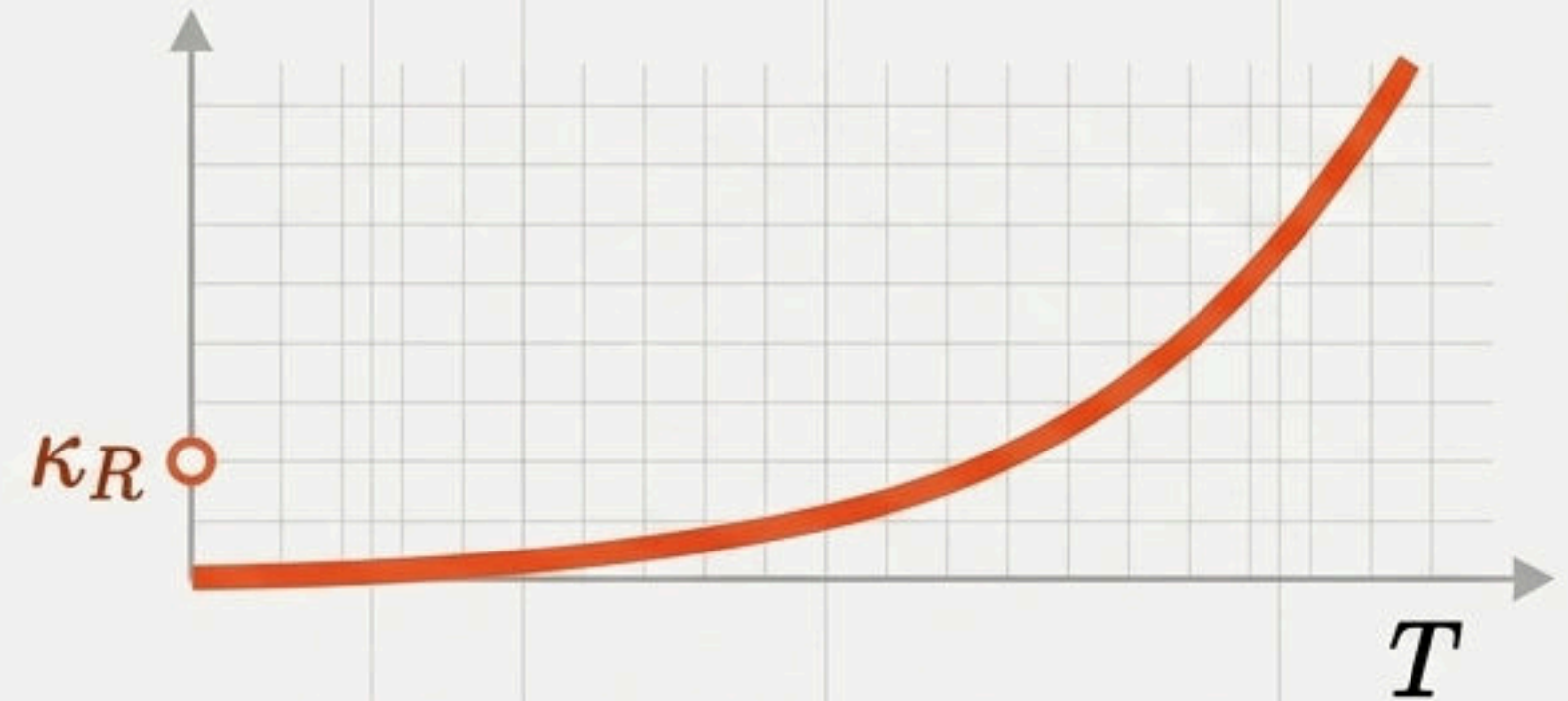
Maintenance Cost (C_M)



$$C_M(T) = T(\kappa_M + \alpha a^p)$$

Pays fixed maintenance overhead (κ_M) every step. Growth is linear.

Repair Cost (C_R)



$$C_R(T) = \kappa_R + \alpha(a/\delta)^p((1+\delta)^T - 1)^p$$

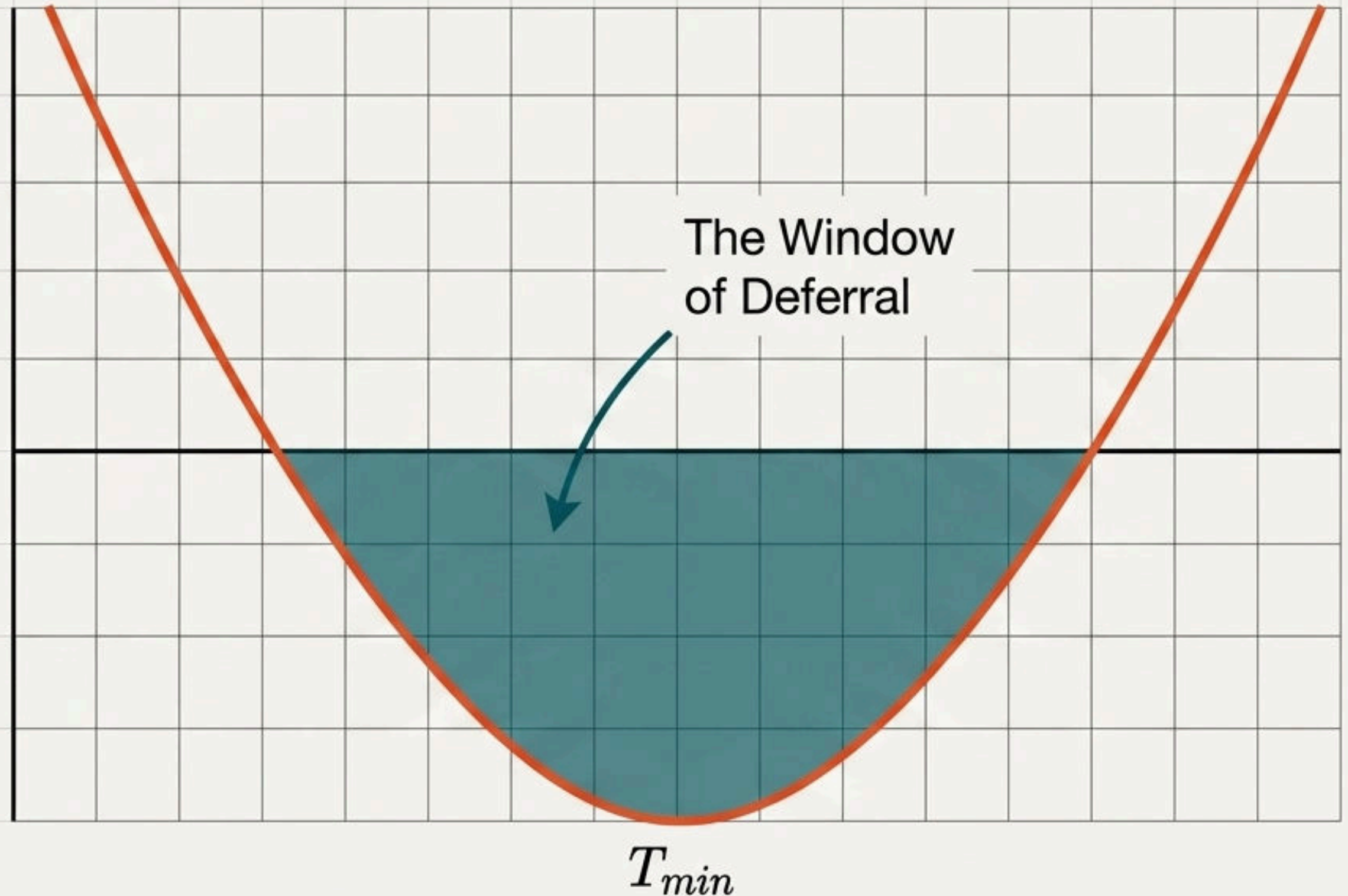
Pays fixed repair overhead (κ_R) once. Growth is convex ($p > 1$).

C_R always defers correction to a single convex-cost operation at the end of the horizon T .

The convex differential.

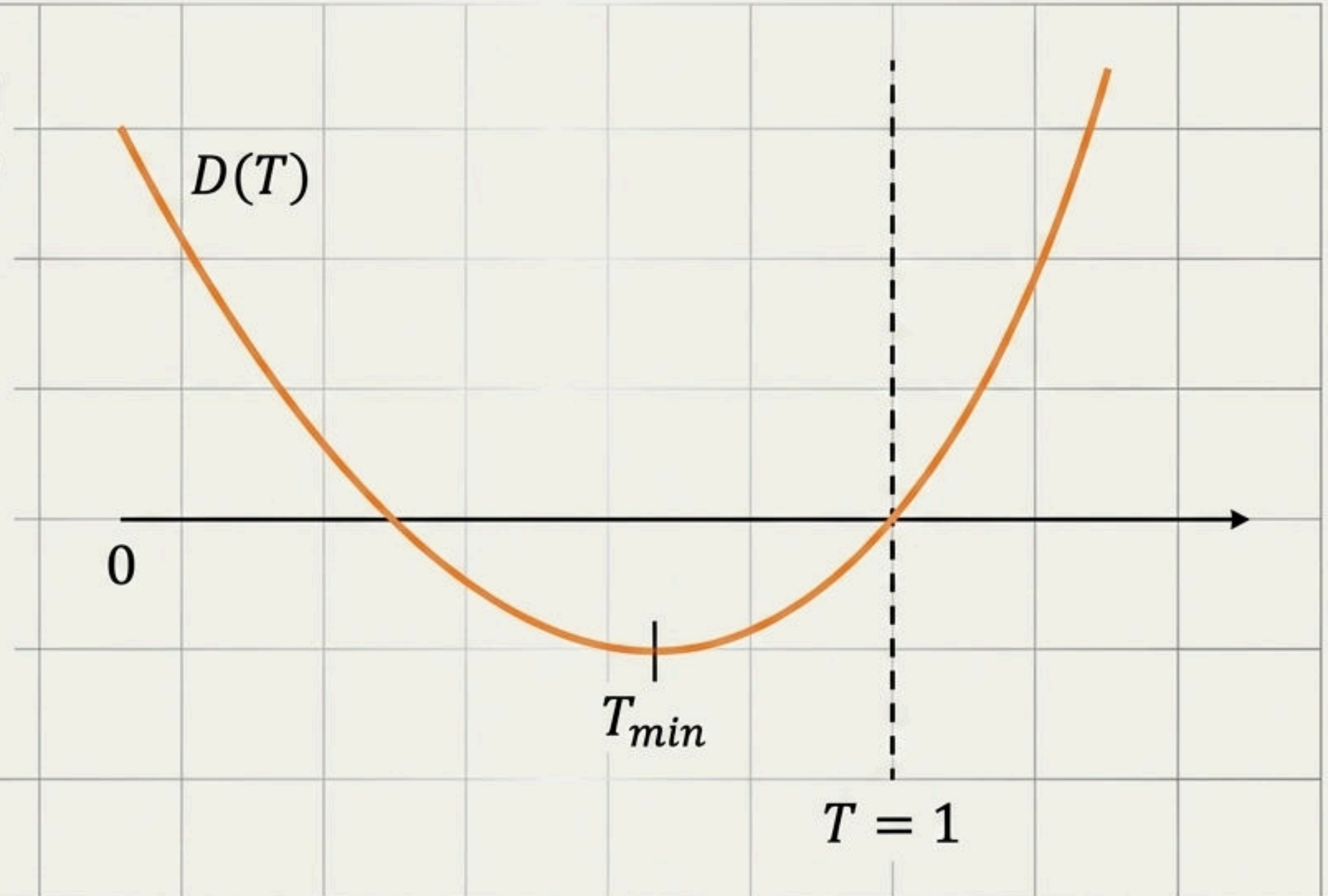
$$D(T) = C_R(T) - C_M(T)$$

- Because C_R is convex and C_M is affine, their difference $D(T)$ is strictly convex.
- A convex function crossing a fixed level can have at most two roots. The roots straddle the unconstrained minimum T_{min} .
- Where $D(T) < 0$, deferred repair is the economically rational choice.



The Early Recovery Condition ($T_{min} \leq 1$)

- Threshold existence is decided in the region $T \geq 1$. If $T_{min} \leq 1$, the system has already passed the point of maximum relative advantage for deferred repair by the time the shortest possible horizon is reached.
- The curve is already on its ascending branch. Delaying further yields rapidly diminishing returns.

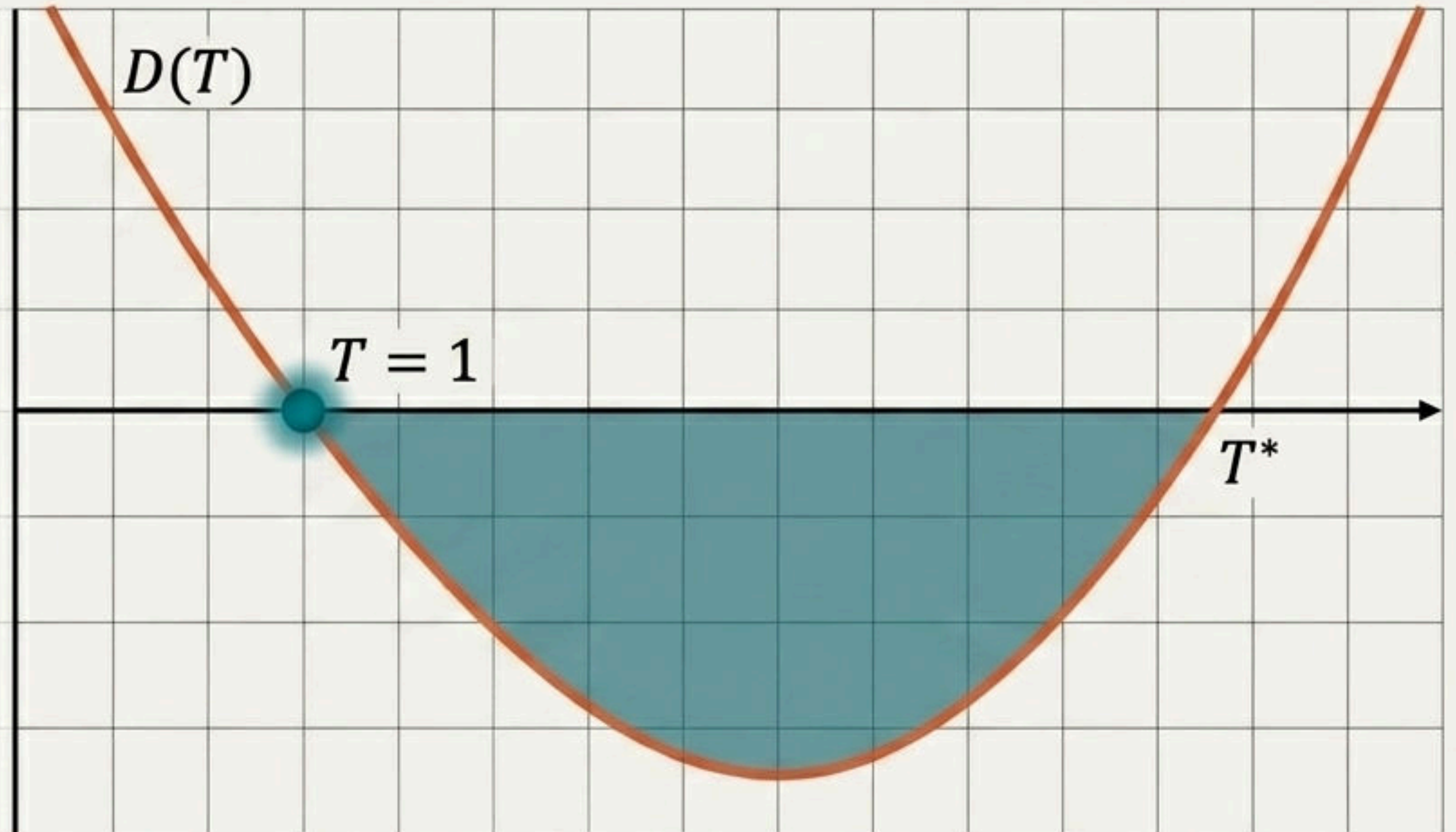


Regime 1: The Immediate Threshold.

The Mathematical Collapse:

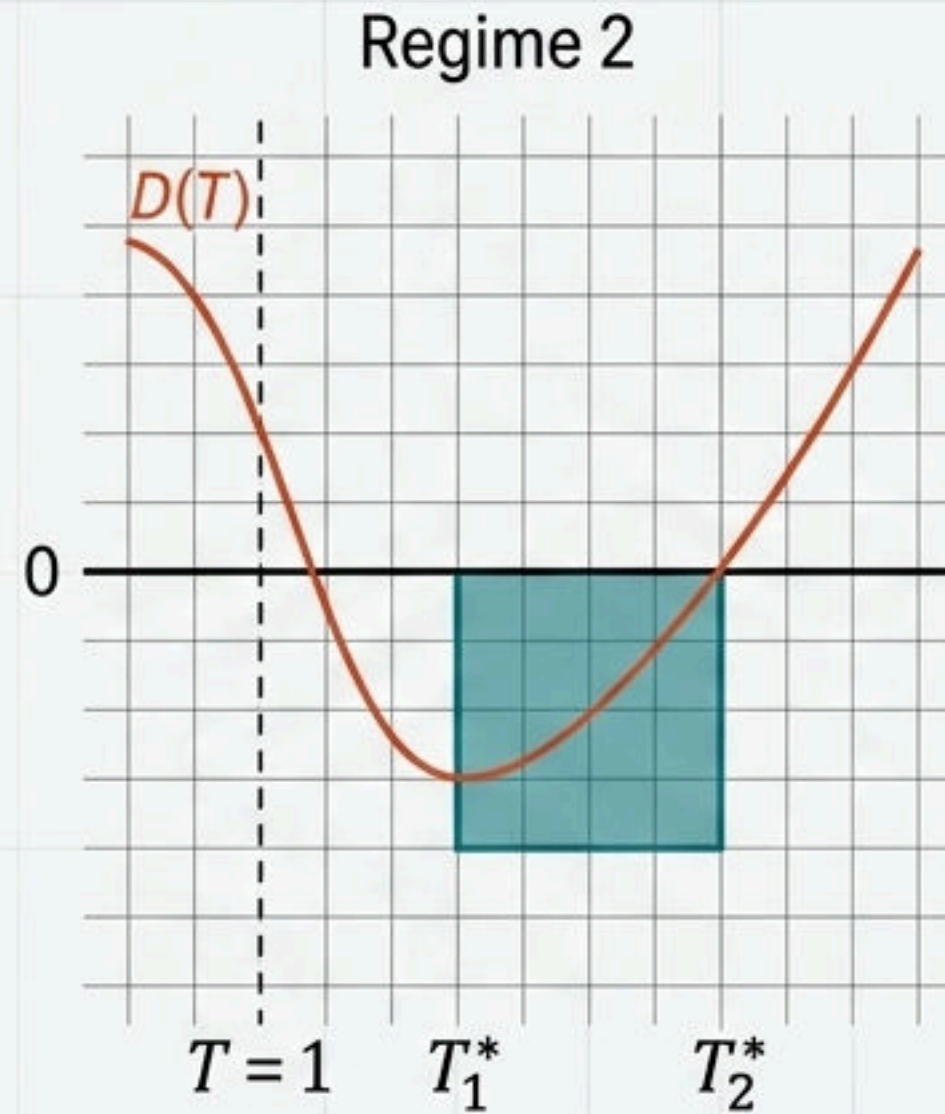
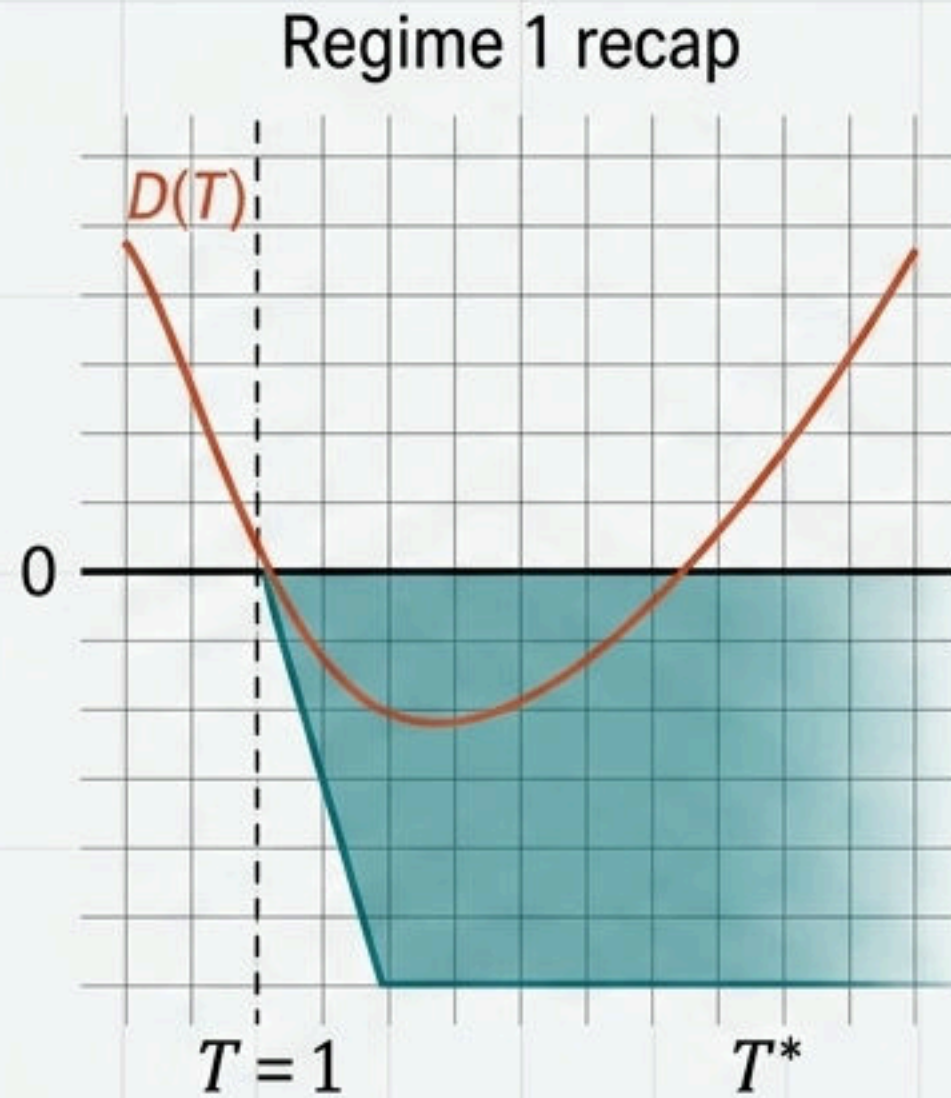
Evaluate D at $T = 1$. Because $((1 + \delta)^1 - 1)^p = \delta^p$, dependencies on a, α, δ, p collapse perfectly.

$$D(1) = \kappa_R - \kappa_M$$



Core Insight: If repair overhead is cheaper than maintenance overhead ($\kappa_R < \kappa_M$), a deferral window opens immediately at $T = 1$. This is purely an overhead-driven phenomenon, completely independent of curvature or compounding.

Regime 2: The Late-Recovery Window.



Condition: $\kappa_R \geq \kappa_M$ (High repair overhead), but $T_{min} > 1$ (Early Recovery fails).

The Geometric Threshold: Instead of comparing κ_R to κ_M , we compare κ_R to κ_R^{crit} —a threshold derived from where the compounding curve's own minimum sits.

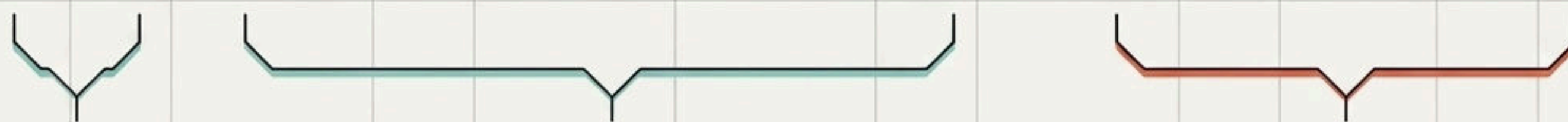
If $\kappa_R < \kappa_R^{crit}$, a deferral window opens later, but it is a finite interval (T_1^*, T_2^*) . Deferral that becomes attractive late does not stay attractive indefinitely.

The Deferral Classification Matrix.

	Column 1: $\kappa_R < \kappa_M$ (Low Repair Overhead)	Column 2: $\kappa_R \geq \kappa_M$ (High Repair Overhead)
Row 1: $x^* \leq \delta$ (Early Recovery)		Early-recovery, no threshold Favorable set: $\emptyset$ Maintenance weakly dominates from start.
Row 2: $x^* > \delta$ (Late Recovery)	Immediate Threshold Favorable set: $[1, T^*)$ Permanent cutover.	Late-recovery, window (if $\kappa_R < \kappa_R^{crit}$) Favorable set: (T_1^*, T_2^*) Finite interval. Late-recovery, no window (if $\kappa_R \geq \kappa_R^{crit}$) Favorable set: $\emptyset$

This classification is exhaustive and exact. The first threshold is governed by overheads; the second by geometry.

Asymptotic Anatomy of the Threshold.

$$T^* \approx T_1 - \frac{\kappa_R}{(p-1)(\kappa_M + \alpha a^p)} - \delta \frac{pT_1(T_1 - 1)}{2(p-1)}$$


Baseline Deferral.

The pure-curvature tolerance of the system when $\delta \rightarrow 0$.

Overhead Penalty.

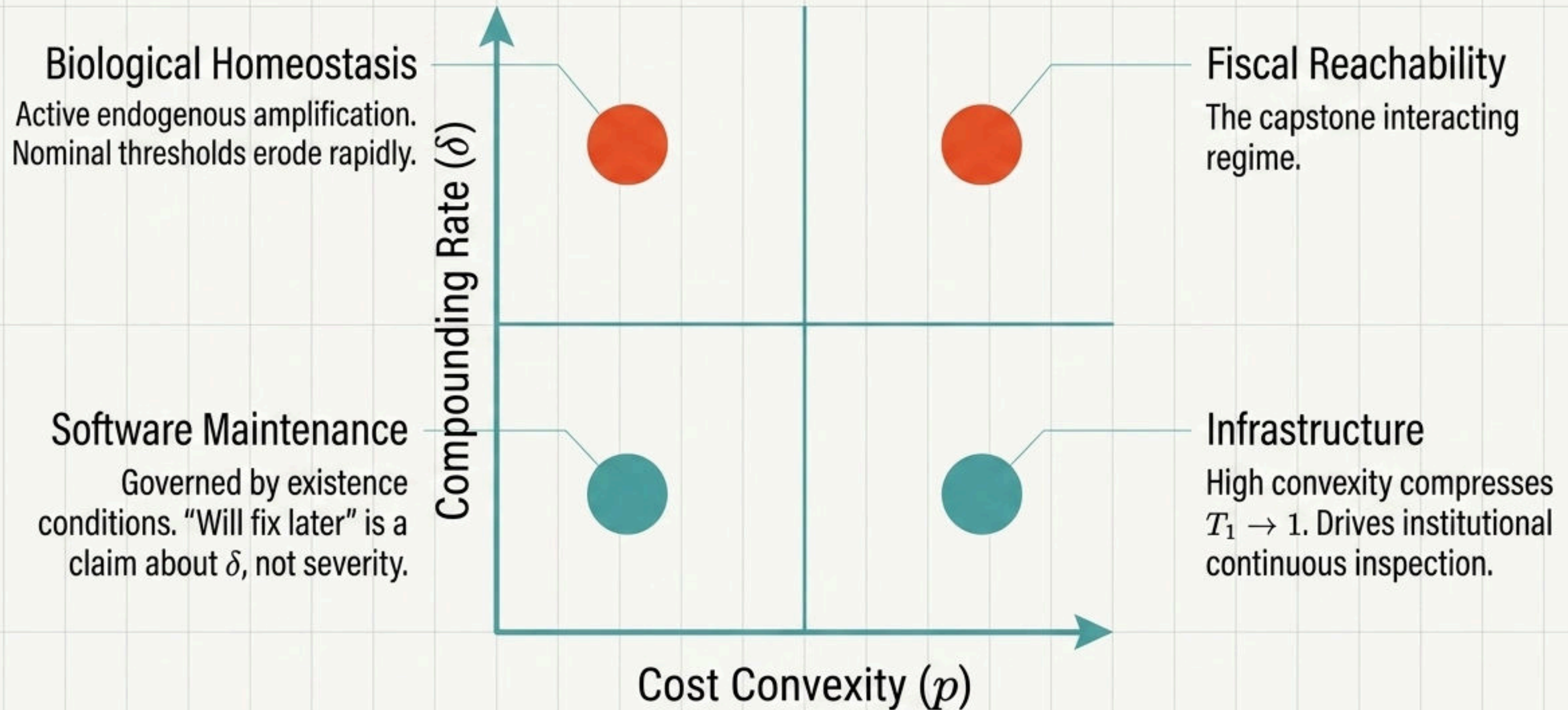
Cheap-to-trigger repair pulls the threshold down.

Compounding Penalty.

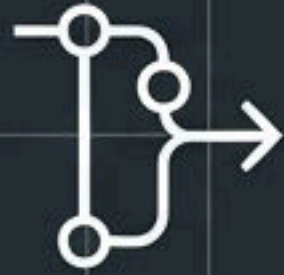
Note that it scales with $T_1(T_1 - 1)$.

A system with a large baseline tolerance (T_1) loses its deferral window fastest under compounding. Tolerance is eroded most rapidly in the systems that appear, on a static reading, most able to afford it.

The Parameter Landscape.



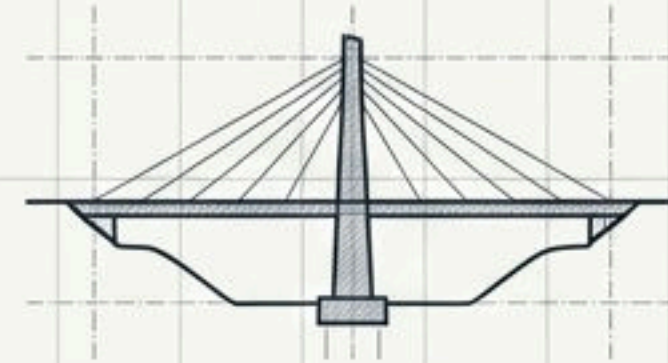
Parameter Extremes: Overhead vs. Convexity.



$$\delta \approx 0$$

Software

Governed entirely by overheads.
If repair initiation (κ_R , shipping a hotfix) is cheaper than continuous monitoring (κ_M , CI overhead), batching minor defects is mathematically optimal.

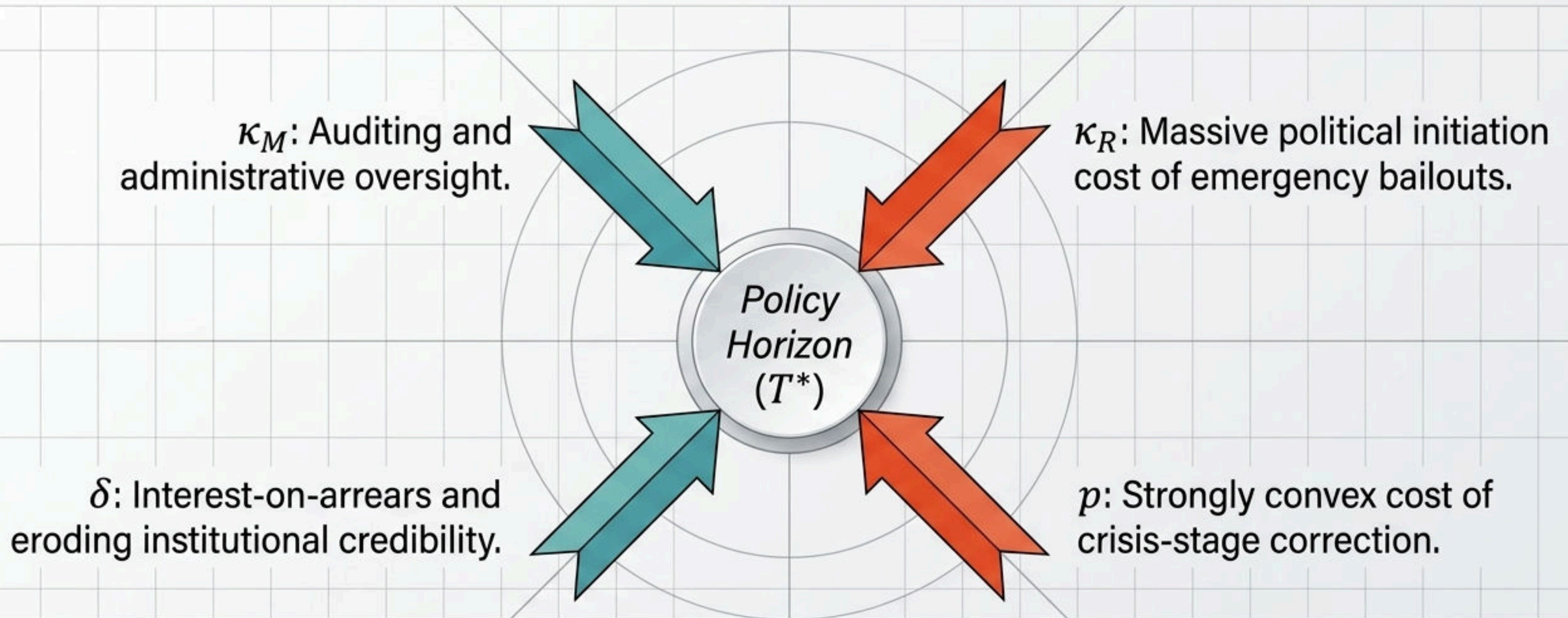


Large p

Infrastructure

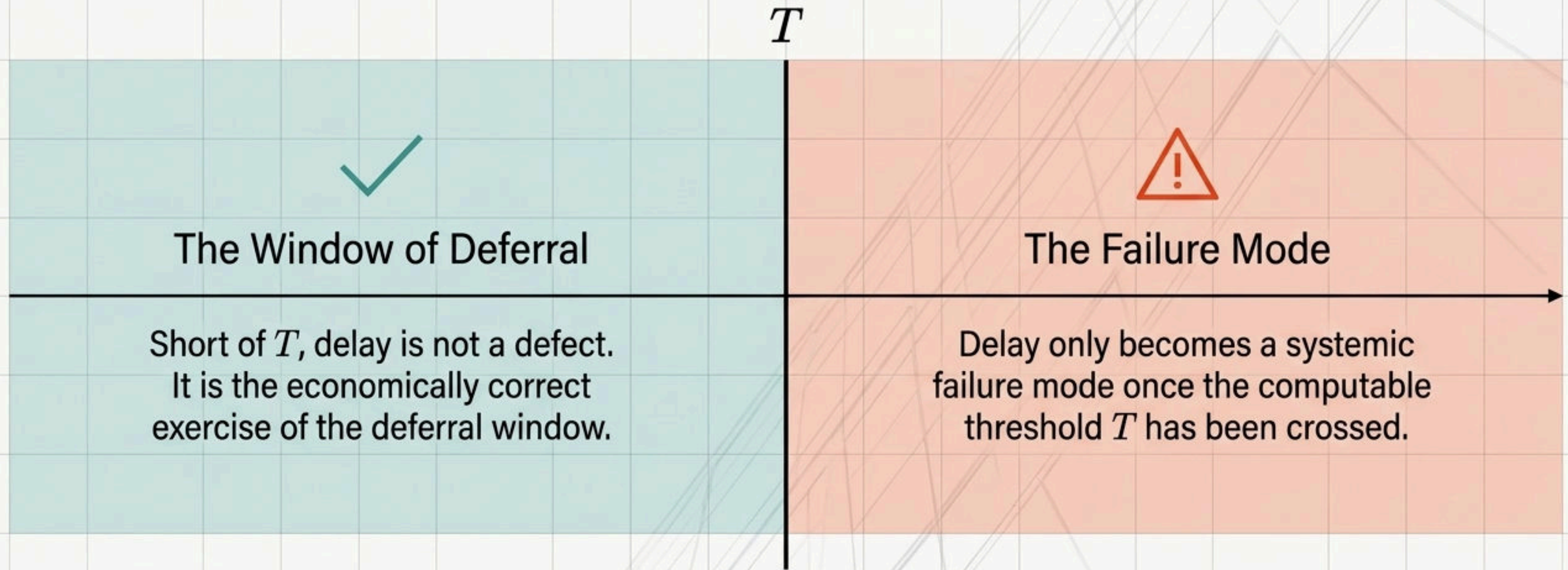
Strongly convex repair costs disproportionately punish accumulated deviation. Large p compresses the baseline threshold T_1 toward 1, requiring continuous correction regardless of how slow individual defects grow.

The Capstone: Fiscal Reachability.



In a fiscal system, T^* represents the **policy horizon**—the point where continuous stabilization becomes strictly cheaper than deferred intervention. Institutional credibility gives a false appearance of high tolerance (T_1) that **structural drift** (δ) rapidly destroys.

Redefining “Repair Delay”.



**Rational system design does not eliminate delay;
it precisely calculates its mathematical boundary.**